such as an empty region of space, the uncertainty principle tells us that from a microscopic vantage point there is a tremendous amount of activity. And this activity gets increasingly agitated on ever smaller distance and time scales.

Quantum accounting is essential to understand this. We saw in the preceding chapter that just as you might temporarily borrow money to overcome an important financial obstacle, a particle such as an electron can temporarily borrow energy to overcome a literal physical barrier. This is true. But quantum mechanics forces us to take the analogy one important step further. Imagine someone who is a compulsive borrower and goes from friend to friend asking for money. The shorter the time for which a friend can lend him money, the larger the loan he seeks. Borrow and return, borrow and return—over and over again with unflagging intensity he takes in money only to give it back in short order. Like stock prices on a wild, roller-coaster day on Wall Street, the amount of money the compulsive borrower possesses at any given moment goes through extreme fluctuations, but when all is said and done, an accounting of his finances shows that he is no better off than when he began.

Heisenberg's uncertainty principle asserts that a similar frantic shifting back and forth of energy and momentum is occurring perpetually in the universe on microscopic distance and time intervals. Even in an empty region of space—inside an empty box, for example—the uncertainty principle says that the energy and momentum are uncertain: They fluctuate between extremes that get larger as the size of the box and the time scale over which it is examined get smaller and smaller. It's as if the region of space inside the box is a compulsive "borrower" of energy and momentum, constantly extracting "loans" from the universe and subsequently "paying" them back. But what participates in these exchanges in, for instance, a quiet empty region of space? Everything. Literally. Energy (and momentum as well) is the ultimate convertible currency. $E = mc^2$ tells us that energy can be turned into matter and vice versa. Thus if an energy fluctuation is big enough it can momentarily cause, for instance, an electron and its antimatter companion the positron to erupt into existence, even if the region was initially empty! Since this energy must be quickly repaid, these particles will annihilate one another after